

Cost Estimation & Summary (INR): Meeting Transcription & Alerting System

This document provides a detailed breakdown of operational costs in **Indian Rupees (INR)** for an application using **OpenAI Whisper API** (Speech-to-Text), **Google Gemini API** (Transcript Processing & Action Item Extraction), and **WhatsApp Cloud API** (Alerts/Notifications).

Exchange Rate Assumption: 1 USD = ₹95.00

1. Unit Pricing Overview (in INR)

Service	Component / Model	Pricing Unit	Base Cost (INR)
OpenAI Whisper	whisper-1	Per Minute	₹0.57 (\$0.006)
Google Gemini (Flash)	gemini-1.5-flash	Per 1M Input Tokens	₹7.13 (\$0.075)
		Per 1M Output Tokens	₹28.50 (\$0.300)
Google Gemini (Pro)	gemini-1.5-pro	Per 1M Input Tokens	₹118.75 (\$1.250)
		Per 1M Output Tokens	₹475.00 (\$5.000)
WhatsApp Business	Utility Template (India)		₹0.11
	Utility Template (USA)	Per Message	~₹1.43 (\$0.015)
	Utility Template (UK/EU)	Per Message	~₹1.90 (\$0.020)
	Twilio		+₹0.48 (\$0.005)
	Surcharge (if using BSP)		

2. Assumptions & Metric Conversions

To calculate costs per hour of meeting transcription: * **Speech Speed:** ~150 words per minute (WPM). * **Word to Token Ratio:** 1 word ≈ 1.33 tokens. * **1 Hour of Audio:** * 60 minutes. * ~9,000 words. * ~12,000 tokens of raw transcript. * **Prompt Overhead:** System instructions, historical context, and JSON schemas add ~1,000 input tokens. * **Total Input Tokens:** **13,000** per hour of meeting. * **Output Size:** Meeting summary, action items, and escalation detection output ≈ 1,500 tokens. * **Total Output Tokens:** **1,500** per hour of meeting. * **WhatsApp Alerts:** Assume **5 utility alerts** are dispatched per meeting.

3. Cost Per Meeting Hour (Itemized)

Below is the cost breakdown for transcribing and processing exactly **1 hour of meeting audio**.

Option A: Standard Stack (whisper-1 + gemini-1.5-flash + India WhatsApp Alerts)

- 1. **Transcription:** 60 mins × ₹0.57 = **₹34.20**
 - 2. **Gemini Flash Processing:**
 - Input: 13,000 tokens × (₹7.13 / 1,000,000) = ₹0.09269
 - Output: 1,500 tokens × (₹28.50 / 1,000,000) = ₹0.04275
 - Gemini Total = **₹0.1354**
 - 3. **WhatsApp Alerts:** 5 alerts × ₹0.11 = **₹0.5500**
- **Total Cost Per Hour (Standard): ₹34.89** (approx. ₹35)

Option B: Premium Stack (whisper-1 + gemini-1.5-pro + India WhatsApp Alerts)

- 1. **Transcription:** 60 mins × ₹0.57 = **₹34.20**
 - 2. **Gemini Pro Processing:**
 - Input: 13,000 tokens × (₹118.75 / 1,000,000) = ₹1.5438
 - Output: 1,500 tokens × (₹475.00 / 1,000,000) = ₹0.7125
 - Gemini Total = **₹2.2563**
 - 3. **WhatsApp Alerts:** 5 alerts × ₹0.11 = **₹0.5500**
- **Total Cost Per Hour (Premium): ₹37.01** (approx. ₹37)

4. Cost Scenarios by Scale

Assuming standard Indian rates for WhatsApp and Gemini Flash.

Scenario 1: Small Startup / Team

- **Volume:** 10 meetings/month (approx. 10 hours of audio).
- **WhatsApp:** 50 alerts sent.

Component	Cost Calculation	Monthly Total
Whisper STT	600 mins × ₹0.57	₹342.00
Gemini 1.5 Flash	10 × ₹0.1354	₹1.35
WhatsApp API	50 × ₹0.11	₹5.50
Total Cost		₹348.85 / month

Scenario 2: Mid-Size Organization

- **Volume:** 150 meetings/month (approx. 150 hours of audio).
- **WhatsApp:** 750 alerts sent.

Component	Cost Calculation	Monthly Total
Whisper STT	9,000 mins × ₹0.57	₹5,130.00
Gemini 1.5 Flash	150 × ₹0.1354	₹20.31
WhatsApp API	750 × ₹0.11	₹82.50
Total Cost		₹5,232.81 / month

Scenario 3: Large Enterprise

- **Volume:** 1,000 meetings/month (approx. 1,000 hours of audio).
- **WhatsApp:** 5,000 alerts sent.

Component	Cost Calculation	Monthly Total
Whisper STT	60,000 mins × ₹0.57	₹34,200.00
Gemini 1.5 Flash	1,000 × ₹0.1354	₹135.40
WhatsApp API	5,000 × ₹0.11	₹550.00
Total Cost		₹34,885.40 / month

Scenario 4: Specific Scale (3,600 Minutes / 60 Hours per Month)

- **Volume:** 3,600 minutes/month (60 meetings of 1 hour each).
- **WhatsApp:** 300 alerts sent (5 alerts per hour).

Standard Stack (whisper-1 + gemini-1.5-flash)

- **Whisper STT:** 3,600 mins × ₹0.57 = **₹2,052.00**
- **Gemini 1.5 Flash:** 60 meetings × ₹0.1354 = **₹8.13**
- **WhatsApp API (India):** 300 alerts × ₹0.11 = **₹33.00** (or ₹429.00 if sending to US numbers at ₹1.43/msg)
- **Total Cost (India Alerts): ₹2,093.13 / month**
- **Total Cost (US Alerts): ₹2,489.13 / month**

Premium Stack (whisper-1 + gemini-1.5-pro)

- **Whisper STT:** 3,600 mins × ₹0.57 = **₹2,052.00**
- **Gemini 1.5 Pro:** 60 meetings × ₹2.2563 = **₹135.38**
- **WhatsApp API (India):** 300 alerts × ₹0.11 = **₹33.00** (or ₹429.00 if sending to US numbers at ₹1.43/msg)
- **Total Cost (India Alerts): ₹2,220.38 / month**
- **Total Cost (US Alerts): ₹2,616.38 / month**

5. Cost Optimization Strategies

[!TIP] 1. **Use Gemini Context Caching:** Caching static system instructions/schemas reduces Gemini input token cost up to 90%. 2. **Leverage Service Windows for WhatsApp:** Responses within 24 hours of customer initiation are free of Meta message charges. 3. **Locational Routing:** Keep using Indian sender IDs and recipient phone numbers to stay at the lowest tier rate (₹0.11/msg) instead of international rates (up to ~₹1.90/msg). 4. **Self-hosted Whisper:** For massive volume (>1,000 hours/month), deploying Whisper on local/cloud GPUs caps STT cost to raw compute/electricity costs.